



Hypothesis Testing

Lesson 4 — Statistics for Data Science

H_0 & H_1 • Type I & II Errors • Alpha • P-value • Z-test • T-test • Chi-square

Null vs Alternative Hypothesis

Every test starts with two competing claims about the population

H_0 — Null Hypothesis

The default assumption.

It states that nothing special is happening — no effect, no difference, no change. We assume it is true until the data gives us strong enough evidence to reject it.

Think of it like a court trial: the defendant is innocent (H_0) until proven guilty.

Examples:

- The new drug has no effect
- The two groups have equal means
- The feature has no impact on sales

H_1 — Alternative Hypothesis

What we are trying to prove.

It states that there IS an effect, a difference, or a change. We collect data to see if the evidence is strong enough to support it.

It is the claim the researcher is investigating — the interesting result they are looking for.

Examples:

- The new drug reduces blood pressure
- Group A earns more than Group B
- The new feature increases sales

Writing H_0 and H_1 Correctly

The null always contains equality. The alternative contains the direction of the claim.

Scenario	H_0 (Null)	H_1 (Alternative)	Type
Drug test	The drug has no effect on blood pressure ($\mu = \mu_0$)	The drug reduces blood pressure ($\mu < \mu_0$)	One
A/B test	Version A and Version B have equal conversion rates ($p_1 = p_2$)	The conversion rates are different ($p_1 \neq p_2$)	Two
Salary gap	Male and female employees earn the same on average ($\mu_1 = \mu_2$)	There is a salary difference ($\mu_1 \neq \mu_2$)	Two
Fraud model	The model performs no better than random chance (accuracy = 0.5)	The model performs better than chance (accuracy > 0.5)	One

Rule: H_0 always contains = (equal). H_1 contains $\neq$ (two-tailed) or $<$ or $>$ (one-tailed).

The Two Types of Errors

We can never be 100% certain — two mistakes are always possible

	H_0 is actually TRUE	H_0 is actually FALSE
Reject H_0	TYPE I ERROR <i>False Positive</i> We rejected H_0 but it was actually true. Probability = α (significance level)	CORRECT <i>True Positive (Power)</i> We correctly rejected H_0 . Probability = $1 - \beta$ (statistical power)
Fail to Reject H_0	CORRECT <i>True Negative</i> We correctly kept H_0 . Probability = $1 - \alpha$	TYPE II ERROR <i>False Negative</i> We kept H_0 but it was actually false. Probability = β (beta)

Trade-off: reducing α (fewer Type I errors) increases β (more Type II errors). You cannot eliminate both simultaneously.

Significance Level (α) and P-value

Two numbers that drive the decision in every hypothesis test

Significance Level α (Alpha)

Alpha is the threshold you set BEFORE running the test. It is the maximum probability of a Type I error you are willing to accept.

Standard values:

$$\alpha = 0.05$$

Most common. 5% chance of false positive.

$$\alpha = 0.01$$

Stricter. Used in medical/pharma research.

$$\alpha = 0.10$$

More lenient. Used in exploratory work.

Alpha is chosen before looking at the data. Choosing it after is called p-hacking and it is a serious problem in research.

P-value

The p-value is the probability of getting results as extreme as yours, assuming H_0 is true.

Decision rule:

$$p < \alpha$$

Reject $H_0 \rightarrow$ result is statistically significant

$$p \geq \alpha$$

Fail to reject $H_0 \rightarrow$ not enough evidence

Common misconceptions:

- ✗ p-value is NOT the probability that H_0 is true
- ✗ $p < 0.05$ does NOT mean a large or important effect
- ✓ It only measures evidence against H_0

The 6-Step Hypothesis Testing Process

Every test — regardless of type — follows this same structure

1**State H_0 and H_1**

Define the null and alternative hypotheses clearly before collecting any data.

**2****Choose α**

Set your significance level. Usually 0.05. This is your decision threshold.

3**Collect data & calculate test statistic**

Gather your sample. Compute Z, t, or χ^2 depending on the test type.

**4****Find the p-value**

The probability of getting your result (or more extreme) if H_0 were true.

5**Make the decision**

If $p < \alpha \rightarrow$ reject H_0 . If $p \geq \alpha \rightarrow$ fail to reject H_0 .

**6****Interpret in context**

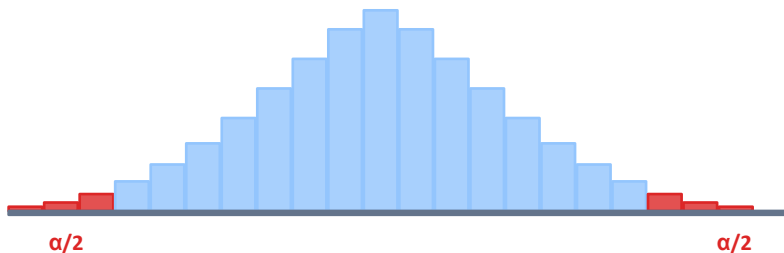
Translate the statistical result into a real-world conclusion. Numbers alone mean nothing.

The next slides apply all 6 steps to real examples using Z-test, T-test, and Chi-square.

One-tailed vs Two-tailed Tests

The direction of your H_1 determines which tail(s) of the distribution you look at

Two-tailed Test ($H_1: \neq$)



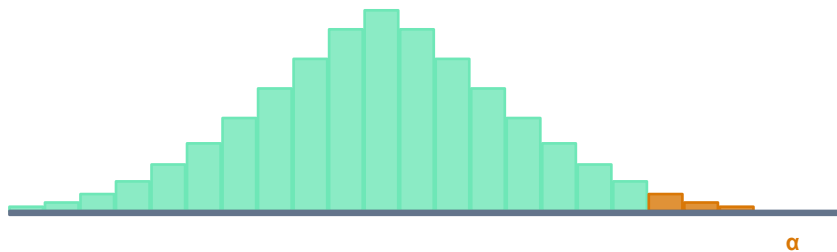
Use when: you want to detect a difference in EITHER direction.

$$H_1: \mu_1 \neq \mu_2$$

Critical region split equally between both tails.
Each tail gets $\alpha/2$.

Example: Is there any salary difference between departments?

One-tailed Test ($H_1: > \text{ or } <$)



Use when: you want to detect a difference in a SPECIFIC direction.

$$H_1: \mu > \mu_0 \quad \text{or} \quad \mu < \mu_0$$

All of α goes into one tail.
More powerful for detecting that specific direction.

Example: Does the new drug REDUCE (not just change) blood pressure?

Z-test

Use when $n \geq 30$ and the population standard deviation is known

Formula & When to Use

$$Z = (\bar{X} - \mu_0) / (\sigma / \sqrt{n})$$

$\bar{X}$ = sample mean

μ_0 = hypothesized population mean

σ = known population standard deviation

n = sample size

Use Z-test when:

- $n \geq 30$ (large sample)
- σ (population std dev) is known
- Data is approximately normal

Worked Example

Scenario: A factory claims average product weight is 500g ($\sigma = 20$ g). You sample 40 products and get $\bar{X} = 492$ g. Is the machine underfilling?

$H_0:$ $\mu = 500$ g

$H_1:$ $\mu < 500$ g (one-tailed)

$\alpha:$ 0.05

$Z = (492 - 500) / (20 / \sqrt{40}) = -2.53$

p-value: 0.0057 (from Z-table)

Decision: $p < 0.05 \rightarrow$ Reject H_0

Conclusion: Strong evidence the machine is underfilling.

T-test

Use when $n < 30$ or population standard deviation is unknown

One-Sample T-test

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$$

When to use:

Compare a sample mean to a known value.

Example:

Is the average customer age 35? You survey 25 customers and get $\bar{X} = 38$.

Two-Sample T-test

$$t = (\bar{x}_1 - \bar{x}_2) / SE$$

When to use:

Compare means of two independent groups.

Example:

Does Group A earn more than Group B? Two separate samples of employees.

Paired T-test

$$t = \bar{d} / (s^d / \sqrt{n})$$

When to use:

Same subjects measured twice (before/after).

Example:

Did training improve scores? Same 20 employees tested before and after.

Key difference from Z-test: uses the t-distribution (heavier tails) because we estimate σ from the sample. As n grows, t-distribution approaches normal.

Chi-square Test (χ^2)

Use when your data is categorical — counts and frequencies, not means

Formula & When to Use

$$\chi^2 = \sum (O - E)^2 / E$$

O = Observed frequency (actual count)

E = Expected frequency (what we would expect if H_0 is true)

Use chi-square when:

- Data is categorical (labels, groups)
- You are working with counts/frequencies
- Testing if two categorical variables are independent
- Testing if observed distribution matches expected

Cannot be used with continuous data or means.

Worked Example

Scenario: Does buying preference (Online vs In-store) depend on age group? Survey of 200 customers.

	Online	In-store	Total
18–35	70	30	100
36–60	40	60	100
H ₀ :	Buying preference is independent of age		
H ₁ :	They are related		
χ ² =	16.16 (calculated from table)		
p-value:	< 0.001		
Decision:	Reject H ₀ — they are related		
Young customers prefer online, older customers prefer in-store.			

Choosing the Right Test

A quick decision guide

Test	Data Type	Groups	Sample Size	σ Known?
Z-test	Numerical	1 group vs known value	$n \geq 30$	Known
One-sample T-test	Numerical	1 group vs known value	$n < 30$	Unknown
Two-sample T-test	Numerical	2 independent groups	Any	Unknown
Paired T-test	Numerical	Same group — before/after	Any	Unknown
Chi-square	Categorical	2+ categories	$n \geq 5$ per cell	N/A

The most important question: Is your outcome numerical (use T or Z) or categorical (use Chi-square)?

Lesson 4 — Key Takeaways



H_0 is the default (no effect). H_1 is what we want to prove. H_0 always contains =



p-value does NOT mean the probability that H_0 is true — a very common mistake



Type I error: rejecting H_0 when it is true (false positive). Probability = α



Two-tailed: testing for any difference ($\neq$). One-tailed: testing a specific direction ($<$ or $>$)



Type II error: keeping H_0 when it is false (false negative). Probability = β



Z-test: large sample ($n \geq 30$), σ known. T-test: small sample or σ unknown



Alpha (α) is set before the test. It is the threshold for making a decision. Usually 0.05



Paired T-test: same subjects before and after. Two-sample: two independent groups



p-value: probability of results this extreme if H_0 were true. $p < \alpha \rightarrow$ reject H_0



Chi-square: categorical data. Tests if two categorical variables are independent